CODE ALPHA

CYBER SECURITY INTERSHIP

TASK 01: BASIC NETWORK SNIFFER

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TASK 01 REPORT

➢ Introduction:

The Basic Network Sniffer project is a Python-based application designed to capture and analyse network traffic. It provides insight into packet structures, protocols, and data flow.

➢ Objective:

1. Capture live network traffic using Python.
2. Display key packet information: Source/Destination IP, Protocol, Ports, Payload.

➢ Tools & Technologies:

Tool/Library	Purpose
Python 3.12	Programming language
Scapy	Packet sniffing and analysis
Tkinter	GUI creation
Threading	Run sniffing in background
ScrolledText	Scrollable display for packets

➢ Implementation Details:

1. Uses Scapy's sniff() function to capture live network traffic in real-time.
2. Performs Layer 3 packet capture for Windows compatibility.
3. Captures TCP, UDP, and other IP packets while extracting key details like IPs, ports, and payloads.

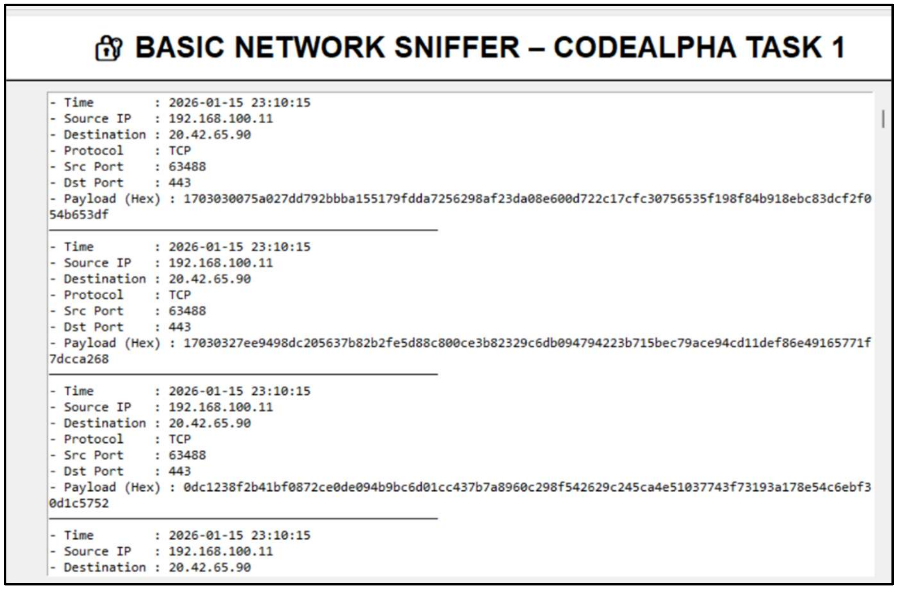
➢ Key Learnings:

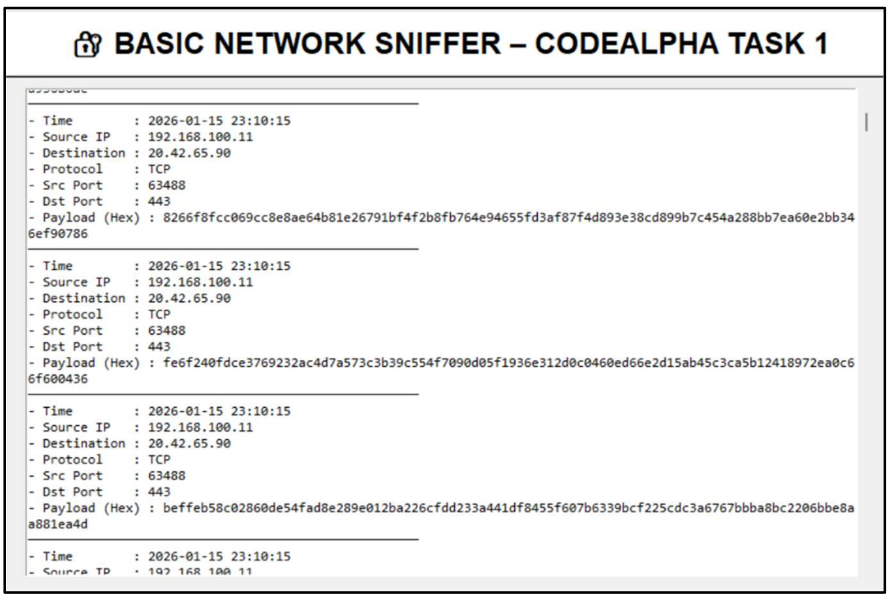
- ➢ Understanding of network packet structure.
- ➢ Real-time packet capture and analysis.
- ➢ Windows compatible Layer 3 sniffing.

➢ Conclusion:

The Basic Network Sniffer successfully captures and analyses live network traffic, providing detailed insights into packet structures, protocols, and data flow. Its professional GUI ensures readability and usability, while Windows compatible Layer 3 capture makes it easy to run without additional drivers. This task strengthened understanding of network monitoring, packet analysis, and Python programming.

- **Sample Output:**





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